

The Effect of Level of Intra Industry Competition and the Intensity of Role of Global Investors toward Degree of Capital Market Integration in ASEAN

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Abstract

By the integration of the ASEAN capital markets, global investors need to diversify among industries in the ASEAN capital markets with the consideration that there is no relevancy regarding to diversify among countries. One way to measure the degree of integration in addition to cointegration techniques are already widely used is the correlation. The use of correlation is done not only to show different degree of integration but also intended to identify the determinant of market integration. The first finding of this study is the different degree of integration at the country and industry levels in the ASEAN which reflects the difference in the benefits of international diversification among industries and countries). The second finding as the important result for this study is the effectiveness of the entropy as a proxy for the level of intra industry competition in Singapore. The implication is global investors need to consider the dynamics of competition among firms within the industry in Singapore as the industry risk for conducting their international diversification strategy. The latest finding is Malaysian bourse still protective so that global investors need to be more active penetration by networking through the local brokerages.

Key Words: capital market integration, level of intra industry competition, intensity of role of global investors, inter industry diversification, entropy

JEL Classification E44, F3, F36, G15

1. INTRODUCTION

1.1. Research Background

According to Click and Plummer (2005) and Yu, et al. (2010) integration of stock exchanges in ASEAN can be seen from the openness of the exchange to international capital flows and movements together (comovement) between the index with regional and international stock exchange market. The opening exchanges in ASEAN, according to Bekaert (1995) because of financial liberalization in the decade of the 1990s. While the comovement occurs, according Pretorius (2002), because of increased capital flows from global investors to emerging markets (ASEAN).

The more open the ASEAN capital markets related financial liberalization makes the flow of international capital into the region continues to increase. Woochan and Wei (2002), Kuper and Lestano (2007) and Sula and Willett (2009) states in addition to an increase in capital flows will have implications for the further integration of ASEAN capital markets also indicates the behavior of global investors in international diversification. Their behavior according to Richards (2005) is the tendency to give higher weight (overweight) in emerging markets such as ASEAN as part of a global portfolio. Further to the already integrated

ASEAN capital markets, the benefits of international diversification will be reduced. This is because diversification will succeed if the pattern of movement of the market return of each stock exchange is different. Roll (1992) states reduced benefits of international diversification will encourage global investors to change the diversification among countries to diversify across industries. The success of diversification depends on the amount of capital and the level of knowledge of global investors.

International diversification strategy would be successful if any global investors could find, not only where capital markets are not integrated but also look at what factors will create an integrated capital market. These factors include Global Push Factors (GPF) and the Local Pull Factors (LPF). GPF are the driving forces of the degree of openness (openness level) in the capital market-related dynamics of international capital flows. While the LPF is the driving forces of each form of capital market i.e. volatility, liquidity and industry sectors. According to Bekaert and Harvey (1995) and Baele and Inghelbrecht (2009), global investor needs to consider the above factors in order to obtain optimality guarantees the benefits of international diversification because the level of integration varies according to time (can be referred to as time-varying integration).

The trend level of integration varies according to time (time-varying integration) can be viewed as a form of capital market integration dynamics. According to Hong (2005) and Baele and Inghelbrecht (2009), the dynamics of the level of integration can be seen by the pattern of correlations between stock market index returns of each country that has a rising trend ranging from tranquil conditions (calm) to the volatile conditions (crisis). During global financial crisis of 2008, the level of correlation between stock market indices at the ASEAN increased. The increase in the level of correlation results in an increase in the degree of integration that lowers the level of international diversification benefits. If global investors want to maintain the ASEAN capital markets as part of a global portfolio, the diversification strategy across countries need to be converted into inter industry diversification.

Argumentation dynamics of capital market integration can be associated with the study results Errunza, et al. (1992) which divides the integration of the three levels of perfect integration, mild segmentation and perfect segmentation. Under conditions of perfect integration, barriers to entry for global investors do not exist and vice versa for the perfect segmentation. In the conditions of mild segmentation, the barrier to entry for global investors is not the maximum. Entry barriers in the form of a difference in treatment as quoted by Bae, et al. (2004) and Sun, et al. (2009), a global investor in the China stock market can only buy shares of series B. While shares of series A are only preferred to local investors (Chinese).

Yu, et al. (2010) found the level of capital market integration in ASEAN tend to vary according to time (time-varying). These conditions encourage global investors to continue to observe the performance of their portfolios. Observation is no longer a partner at the state (country) level but has been in the industry pair level as in the Benson and Green (2005). This is because each country in ASEAN will have a unique structure in addition to industry related differences in the level of openness to foreign capital flows. The uniqueness of the industrial structure of competition is reflected in the characteristics of each industry. In order to ensure the success of inter industry diversification, the global investors need to form a portfolio of industrial partners with a low level of competition. This is based on the findings Carrieri, et al. (2004) that the lack of competition resulted in a lower risk of inter industry diversification. With the growing integration of ASEAN capital markets by making time-varying, global investors need to analyze the dynamics of competition in every industry through a measuring instrument with adequate theoretical basis of the entropy index Ruefli (1990). The use of the entropy index is expected to help ensure the success of diversification but also to note the differences in the level of openness in each ASEAN country.

1.2. Gap of Research

According to Cheng (2000) and Yusof and Madjid (2006), it was identified two groups of studies of integration i.e. IRGISG (Intensity of Role of Global Investors Study Group) and Industry Factors Study Group (IFSG). Examples of IRGISG namely Bekaert, et al. (2002), Edison and Warnock (2003) and Froot and Ramadorai (2008) and examples of IFSG are Tu (1998), Faff and Mittoo (2003) and Carrieri, et al. (2004). Both of studies claim that Local Pull Factors (LPF) and Global Push Factors (GPF) are more important to one another, although they refer to the theory of stock market interdependence from Pretorius (2002). These conditions will bring to the first research gap i.e. the divergence from Local Pull Factors and Global Push Factors as main determinant of market integration.

The theory of stock market interdependence of Pretorius (2002) says there are three determinants of the integration such as the contagion, economic integration and capital market characteristics. Capital market characteristics include volatility, liquidity and industry similarity. From the theory of stock market interdependence it is stated that Global Push Factors associated with contagion because contagion occurs as a result of increased international capital flows. Meanwhile, Local Pull Factors are characteristic of the appeal of a capital market for global investors.

In the view of IRGISG, Global Push Factors will be more relevant as a determinant of integration due to the role of global investors since the era of liberalization of capital markets increased. The role of global investors is demonstrated by the increasing global investor fund flows. According to Froot and Ramadorai (2008), the increased flow of funds will affect the global investor in the stock market index of a country. Dvorak (2005) and Aggarwal, et al. (2009) states the foreign funds flow will affect the level of integration for a more open capital markets would be utilized by global investors to be more aggressive in penetrating. But the study of Edison and Warnock (2003) and Bae, et al. (2004) stated that some countries increase the level of protection for local investors. This is because the more negative the dominant role of global investors who are expected to take action to destabilize the local stock exchange to trigger an increase in the volatility of the stock and in turn lead to bubble. Nevertheless the study Bekaert, et al. (2000) declared that global investor also has the positive role that can bring improved performance and liquidity of the market index trading. Which then becomes a problem is the behavior change of global investors is hard to be detected at any time by the regulator.

Meanwhile in the opinion of Industry Factors Study Group (IFSG), Local Pull Factors more relevant as a determinant of integration as an industry sector will have an attraction for global investors. Before the flow of investment funds, global investors will study the characteristics of each industrial sector. Based on the study of MSCI Barra, every country in ASEAN has a unique respective industry. According to the study of Carrieri, et al. (2004) and Dutt and Mihov (2008) the industrial sector is expected to affect the level of integration because it has a risk exposure that is worthy of consideration by every global investor in calculating the benefits of international diversification. In the classical model ICAPM, the higher the expected return required of an industry makes the higher the risk to be borne by the industry. Moreover it would be true if the industry is categorized as the global industry such as the Faff and Mittoo (2003). More relevant Local Pull Factors (LPF) as a determinant of integration as well as the industrial sector has two arguments i.e. the similarity of industrial structure and industry strategic risk. According to Roll (1992), industrial structure similarity is that if two countries have similar industrial structures, the comovement between the two countries in the market index will increase along with the high concentration of cash flow.

While it also industry strategic risk associated with competitive conditions in the industry. These factors should be considered by global investors because it adds to the risk component of international diversification. Menchero and Morozov (2011) declared that the

global investor can further enhance the benefits of diversification through a more focused strategy of diversification in industries with low levels of competition. But the next question arises of how to measure the level of competition. Biker and Haaf (2002), Eun, et al. (2011) and Hsin and Tseng (2012) measure the level of industry competition with HHI (Herfindahl Hirschman Indices), whereas the HHI is designed to industry concentration. Industry concentration may reflect the nature of competition in the long life industry. In the turbulence industry such as information technology, concentration does not reflect the nature of competition in the industry and the consequences it was to be inadequate if HHI is continue to used as a proxy of competition. Thus it will create a second research gap that is the need for measures the level of competition in the industry as a more appropriate because it is generally a qualitative measure of competition. One of example is Porter's Five Forces. So this study takes a measure of competition is more quantitative. Gauge this competition is the entropy index by Ruefli (1990) that will measure how drastic changes in its ranking in the industry for a period of observation. Entropy indices are calculated by OTSA (Ordinal Time Series Analysis) is considered superior to the HHI.

Based on the first and second gap, it will also be created the third gap of this study i.e. how to model simultaneously both determinants of the level of integration of both Local Pull Factors (LPF) and the Global Push Factors (GPF). Simultaneously modeling is expected to justify the theory of stock market interdependence of Pretorius (2002) that these two equally important factors. Simultaneous modeling of both the determinants of integration is still dominated by panel data regression and cointegration techniques. Panel data regression conducted by Hong (2005) and Bekaert, et al. (2011) find that the LPF is more important than GPF in emerging markets (including ASEAN). Cointegration techniques in ASEAN were conducted by Click and Plummers (2005) and Kuper and Lestano (2007) with more focus on Global Push Factors. This is because the motive for their study is the detection of long run equilibrium relationship between ASEAN countries indexes and index of developed countries. The findings of these two approaches are contradictory, so in our opinion it was taken a more comprehensive modeling.

1.3. Problem Formulation and Research Objectives

If the capital markets in the ASEAN have been integrated, the (global) diversification across countries is no longer relevant. The right choice for internationally diversified of global investors is diversified among industries. This is because at the time of the integrated market conditions, the pattern of movement in any stock market return will be the same. Because the pattern of movement in any stock market return as the correlation between the markets return has increased. Based on the classic literature of the Markowitz investment, when the return correlation in the ASEAN market is increasing, the benefits of international diversification for global investors will be reduced. Benefits of international diversification is generally measured by the Sharpe Ratio as on Ratner and Leal (2005) who has managed to find the phenomenon of integration of the industry level in developed markets.

Reduction in the benefits of international diversification as a result of the capital markets in ASEAN has been integrated is certainly a problem for global investors. This is due to a common pattern of movement in any stock market returns that raise the level of systematic risk of the market respectively. But for global investors will be less efficient if they divert the flow of capital out of ASEAN. This would increase the total cost of investment transactions in addition to the more difficult for global investors get higher returns outside the ASEAN which has been more perfect market.

Therefore, global investors remain focused only on ASEAN capital markets; it's just that they now need to better see the condition of the industry. According Menchero and Morozov (2011), emerging markets such as ASEAN still has the potential for a prospective

industry. Industries are considered to have a competitive advantage because it categorized as GICS (Global Industry Classification Standard) which has always been a top priority of global investors. However, due to capital market integration in ASEAN is time-varying and the study of Benson and Green (2005) states that the time-varying level it had reached in the industry makes the global investors need to be constantly up-date portfolio of each pair of industries in the hope of international diversification benefits remain awake.

Thus the main problem of this study is the need for global investors when faced with the situation in the capital markets in ASEAN are increasingly integrated to consider two main factors from diversification across industries namely the level of intra industry competition and the intensity of role of global investors, where previously they also need to analyze the dynamics of the level of integration at the level of related industries. Furthermore from the main problem as stated before, we have three research objectives as follow:

- a) Test whether there are differences in the level of integration at the level of countries and industries to the 5th ASEAN equity markets on the basis of the argument time-varying world market integration from Bekaert and Harvey (1995).
- b) Examine the influence of the level of intra industry competition on the degree of integration for the 5th ASEAN equity markets with reference to an argument of similarity of industrial structure from Roll (1992), Pretorius (2002), Faff and Mittoo (2003), Dutt and Mihov (2008) and strategic industry risk of Ruefli (1990).
- c) Examine the influence of the intensity of the role of global investors on the level of integration in the 5th ASEAN equity markets on the basis of international capital mobility argument from Marston (1995) and Miskhins and Eakins (2000) and the role of global investors from Bekaert and Harvey (2000) and Froot and Ramadorai (2008).

2. LITERATURE REVIEW

2.1. Alternative Measure for Degree of Market Integration

One of the important issues of discussion of capital market integration is a way of measuring the level of integration. Yu, et al. (2010) suggests three ways of measuring the level of integration. The first way that is price-based measure. In this way related to the efforts equate the return of various asset classes of the same risk between countries with different economic conditions. Price-based measure is related to the concept of LOOP (Law of One Price) that will be described by three measurements i.e. the CIP (Covered Interest Rate Parity), UIP (Uncovered Interest Rate Parity) and RIP (Real Interest Rate Parity).

Quantity-based measure as a second way to measure the level of integration will be demonstrated by saving investment correlation as Feldstein and Horioka formulation. While the last way that the regulatory-based measure will be explained by capital controls and stock market comovement.

2.1.1. Price-based Measure

Yu, et al. (2010) states this measure is a condition in which foreign nationals can be traded asset or securities with the local citizens so that the price of assets or securities between the two countries will approach the price of assets or securities in the global market. The more easily the assets or securities transaction is the price of assets or securities referred to be the price in the global market. These conditions encourage the enactment of the LOOP (Law of One Price) which depends on two factors namely the interest rate and currency movements. If the currency is tied together and the nominal interest rate is then used CIP condition. Meanwhile, when the currency is not tied to the UIP condition and the last when the real interest rate is the relevant refer to condition of RIP.

2.1.2. Quantity-based Measure

The idea of saving and investment from Fledstein and Horioka quoted by Yu, et al. (2010) that if the stock market is open then the real interest rate would equalize economic conditions across the entire country so that savings and investment will not be correlated. Conversely, if capital markets tend to be closed due to low mobility of capital flows then the real interest rate will vary according to the conditions of each country. This resulted in increased correlation between saving and investment. The model is as follows:

$$(I/Y)_i = \alpha + \beta (S/Y)_i \quad (2.1)$$

Where I is investment in country i, Y is gross national income (GDP) state I, S is the savings rate in country i. The smaller of β makes the correlation of saving and investing more decrease, which means the more open of capital markets. Otherwise if the greater of β (close to 1) makes the mobility of fund decreases and the more closed of capital markets.

2.1.3. Regulatory-based Measure

According to Bekaert (1995) the application of capital controls is more related to the constraints for the global investor obtained from the IMF Annual Report on Exchange Agreement and Exchange Restriction (AREAER). According to the study of Stulz and Wasserfallen (1995), AREAER was determined by using dummy variables are declared open market conditions, namely 1 and closed is 0. Because of the status of the dummy variable, index of AREAER need to be modified. But Bae, et al. (2004) states capital controls represent over the determinants of the level of integration so that many researchers more preferring a stock market-based comovement as a regulatory measure.

Stock market comovement is a measurement of the level of integration of capital on regional and international level by analyzing the rate of return of each stock market. There are several methods to measure the stock market comovement ranging from the correlation as traditional methods to the more modern methods such as VAR for the short term analysis and cointegration for the long term analysis. In addition it is also developing other methods of the most modern such as Chaos and Synchronization. Chaos is more related to the random pattern (fractal) in the integration while synchronization is related to the pattern of similarity between the movement of stocks in emerging and developed markets.

2.2. Summary of Time-Varying World Market Integration (TVWMI)

Dynamic conditions of integration that occur in the emerging markets makes the perfect integration will never happen. Similarly it was applied to the form of perfect segmentation. When measured by the correlation, the value of the degree of integration according to the concept of Errunza, et al. (1992) should be valued at between 0-1. If the correlation is negative the concept of integration is irrelevant because the global investors still benefit from international diversification. Efforts to secure the benefits of international diversification makes polarization of perfect integration and perfect segmentation is not relevant so just stay mild segmentation. According Errunza and Losq (1985) and Sun, et al. (2009), the mild segmentation will only have restricted investor (investor exposed to the rules of the stock restriction) but still have access to eligible securities (shares of series B special for global investors) because the rules are loose again. In this condition no longer the asset pricing model of ICAPM but CRS (Conditional Regime Switching) which is the core model of Time-Varying World Market Integration (TVWMI).

Some of the properties the model time-varying world market integration from Bekaert and Harvey (1995) can be traced from asset pricing models for integrated perfect conditions with assuming no exchange rate risk. Asset pricing model (conditional CAPM) is as follows:

$$E_{t-1}[r_{i,t}^A] = \lambda_{t-1} \text{cov}_{t-1}[r_{i,t}^A, r_{w,t}] \quad (2.2a)$$

where $E_{t-1}[r_{i,t}^A]$ is the expected excess return on stock A is conditional on the state i , r_w is the return on a portfolio of stocks that formed the world-weighted value, cov_{t-1} is the covariance operator conditional on time $t-1$ and λ_{t-1} is the expected value of the covariance risk of the world's stock portfolio at time $t-1$. Risk free rate is equal to zero because the return is set at $t-1$. While perfectly segmented market conditions and the same assumptions, the asset pricing models are as follows:

$$E_{t-1}[r_{i,t}^A] = \lambda_{i,t-1} \text{cov}_{t-1}[r_{i,t}^A, r_{i,t}] \quad (2.2b)$$

A stock is rated at covariance with the return on market portfolio for country i the r_i and λ_i as the local risk. At the national level, the model (2.2b) becomes:

$$E_{t-1}[r_{i,t}] = \lambda_{i,t-1} \text{var}_{t-1}[r_{i,t}] \quad (2.3)$$

Model (2.2a) and (2.2b) focus on an integrated capital market conditions and perfectly segmented. Suppose that a market will be highly integrated or highly segmented and then when there is a change from segmentation to integration conditions or conditions of integration into segmentation, then the valuation of cash flow and return generating process will change. Transfer (switching) may be total or only partial. To overcome the problem of difficulty of this shift, the CRS is used empirically. Think of it S_{it} is observed that the variable is set at 1 when integrated capital markets and by 2 when the stock market is segmented. In the regime 1, the return will be assessed by the mean of the conditional distribution of the model 2.2a. Being in regime 2, the average conditional return on the model set 2.2b. At one point in time then there is the possibility of regime switching is done by positive switching probabilities. From an econometric point of view with a collection of information Z_{t-1} (usually in the form of lagged dividend yields and lagged ratio of equity market capitalization and GDP), it was formulated the conditional mean of time-varying return for stock A in country i at t such as:

$$E_{t-1}[r_{i,t}^A] = \Phi_{i,t-1} \lambda_{t-1} \text{cov}_{t-1}[r_{i,t}^A, r_{w,t}] + (1 - \Phi_{i,t-1}) \lambda_{i,t-1} \text{var}_{t-1}[r_{i,t}] \quad (2.4)$$

where the parameter $\Phi_{i,t-1}$ will have a value in the interval (0,1) which is a time-varying assessments of econometrics for possibility (likelihood) that market will be fully integrated. It can be interpreted as conditional probability of regime 1, $\Phi_{i,t-1} = \text{prob}[S_t^i = 1 \mid Z_{t-1}]$.

2.3. Stock Market Interdependence

According to Pretorius (2002), the assumption of stock market interdependence is LOOP. LOOP itself states an integrated market is the market where the asset has the same expected return regardless of where assets are traded. As for the capital market is segmented then the expected return of asset markets will depend on the location of the related asset. LOOP is a consequence of the enactment of global investors needs to consider the global risks. LOOP enforceability will depend also on how much the mobility of international

capital flows. Mobility of international capital flows will increase the linkage of each stock market. International capital flows will become into the mechanism of global investors for international diversification activities. On the one hand they will secure the investment in capital markets of host countries and on the other hand they also will secure savings (risk-free investment) in the country of origin. If there is no substitution effect between saving and investment, international capital flows will continue to flow and the stock market will raise interdependences. Pretorius (2002) states independences of stock market will depend on contagion, economic integration and capital market characteristics.

2.3.1. Contagion

Contagion can be described as the comovement of the stock market is not caused by the general movement of fundamental factors. There are two factors that work such as the informational factors and institutional factors. Informational factors based on well-known from the comparison between the stock market and the "Keynesian Beauty Contest". By the same analogy with the "Keynesian Beauty Contest", investors in the stock market will sell its investment in specific asset class if they believe that other investors will sell their investments in the same class. Herding behavior of investors will lead to excessive volatility as noted also by Bekaert, et al. (2005).

While institutional factors related to redemption and about two stage investment strategy of hedge fund. Most of the flow of funds into emerging market is open end fund that commonly purchased by global investors. When faced with large scale withdrawal, then the hedge fund will sell all the assets into more liquid market or they will allocate their assets into several indices-weighting. Their action would create an excessive decline in the performance of the market.

2.3.2. Economic Integration

There are two explanatory factors that is bilateral trade and macroeconomic variables like as interest rate and inflation. The extent of correlation between variables was applied to the two countries thus the correlation between the two countries over the market return will also increase depending on the closeness of the bilateral trade between the two countries. When conducting the return correlation testing in ASEAN, Click and Plummers (2005) finds the similar pattern to the return correlation between Singapore and Malaysia is stronger than the return correlation between Singapore and Philippines.

Bracker and Koch (1999) states that the interest rate and inflation has an influence on the market return. So the correlation between the two variables will also influence the correlation of the market return. It is also reinforced by Roll (1992) that the interest rate and inflation will be considered by global investors when making asset valuations. In contrast to the bilateral trade is positively related to the correlation of stock returns, therefore interest rate and inflation would have a negative influence each other.

2.3.3. Capital Market Characteristics

There are three components of capital market characteristics i.e. the volatility, liquidity and industry sectors (but that overlooked here is the industrial sector). The argument is that when two countries have similar industrial structures, the correlation between the two market indexes will rise. For example, when the two markets in emerging market index is dominated by the stocks in a sector such as Oil & Gas, so when a decline in world oil demand will result in a significant reduction of its share price of Oil & Gas in the two countries. The dominance of this industry sector was much easier to make international factors affecting the movement of market indices compared to the two countries if these countries have a heterogeneous structure of the industry. One important part of the industrial structure

according to the study of Ng (1995) is the industry concentration. Industry concentration relates to the opinions of Roll (1992) about the similarity of industrial structure that is the stock return correlation between two stock market when the composition of the growing industrial sectors of both countries has similarities. Increase in correlation is not only because the flow of funds from bilateral relations on the basis of economic policy and business cycles but also due to the homogenization of the international diversification of industrial structure on a global investor. Homogenization of the industry structure here is the dominance of the industrial sector on a consolidated market index.

Roll (1992) then describes the context of the industrial structure in terms of volatility and correlation difference. When an industrial sector in a country has a high volatility will not necessarily follow the same conditions in other countries. This happens because of the dominance of the industrial sector as the dominant sector in each country will vary according to the economic potential of each country. Volatility that occurred that was caused by the excess of (induced) negative international diversification, which in the beginning but then expect an increase in return that there is an increase in correlation. In a study of Roll (1992) the Herfindahl index was used as a proxy for industrial structure affects the volatility of returns. These results reinforce the view that the industrial structure has the potential to be a correlation of stock returns explanatory variables. This is because the volatility or the variance return is the decisive element return correlations are important in addition to the covariance of two related stock returns.

Meanwhile Arshanapali studies, et al. (1997) and Bekaert and Harvey (1997) suggests the increase in correlation is due to the increased volatility of stock returns. But as soon as they argued that the increased volatility of stock returns will be an attraction for global investors in the international diversification strategy. Despite the high volatility of return raises the level of correlation, but Chen and Zhang (1997) suggests the benefits of international diversification remains a reliable global investors from an industry that provides a higher return. This condition is realized when global investors put forward as a partner country portfolio and this is reinforced Bekaert and Harvey (1997) and Cha and Sekyung (2000) who found a low correlation between the market return the developed and developing countries. One other issue about the relationship between industry sectors with the stock market interdependence is the relationship between industrial sectors with other industrial sectors. Park and Woo (2002) found a correlation significance of the industry return index in developed countries over the period 1973-2001. Of particular interest is the correlation between the level of the industry in general the European countries is higher when compared to USA, except for industrial TMT (Technology, Media and Telecommunication). The findings of Park and Woo (2002) for non TMT industries in line with the assumption of bilateral trade in economic integration. While for the TMT industry showed higher idiosyncratic risk of the TMT (USA) from TMT (Europe) like as IT Bubble in Nasdaq.

3. CONCEPTUAL FRAMEWORK & HYPOTHESIS DEVELOPMENT

3.1. Conceptual Framework

Initially the study group of capital market integration is developed by arguing with the theory LOOP (Law of One Price) i.e. integrated market is a market where assets have the same expected return no matter where the assets are traded. As for the capital markets are segmented then the expected return of these assets will depend on the location at which the asset is traded. Based on return and risk the relationship on an integrated capital market, global market risk is relevant and should be considered by the investors. But when conditions are segmented only relevant domestic risk and should be considered by every investor. The next problem is when the condition of capital markets will experience an integrated and

segmented it has not been identified properly. The trend that the integrated capital markets, especially in emerging markets such as ASEAN has indeed been received. It makes the study group to search for evidence of integration becomes ineffective. Even there are such groups which concern to that by variation of methodology ranging from SVAR, SVEC and Chaos.

Thus more relevant to our study is finding the determinant factor of the integration domain corresponding to another study group. We developed two streams of study groups that seem more critical look at each of its determinant factors. These two groups namely IRGISG (Intensity of Role of Global Investors Study Group) stating the integration of capital markets will be determined by international capital mobility with such arguments assume the role of global investors and conditions UCIRP (Uncovered Interest Rate Parity) should be considered. Therefore come from a country outside the capital markets, the determinant is referred to as the Global Push Factors (GPF). As with the IFSG (Industry factor study group) state that the integration of capital markets will be influenced by the industrial structure and comovement. Bases this argument is the similarity of industrial structure and industry strategic risk. Determinants of IFSG called Local Pull Factors (LPF) because it comes from within the capital markets which became the country comparative advantage for global investors. According to Roll (1992), comparative advantage reflected of an industry sector dominance in related capital markets of such country.

These two determinant factors integral to each other according to the study group of IFSG and IRGISG, then in this study we propose two factors combined with the reason both of them would be just as important for global investors in order to effectiveness of international diversification strategy. This means that when an integrated capital market conditions so global investors need to see the local push factors are very important that the level of intra industry competition. Instead of time segmented capital market conditions so global investors can better optimize the global push factors in the form of increased international capital flows. Then conceptual framework is described at figure 3.1 below:

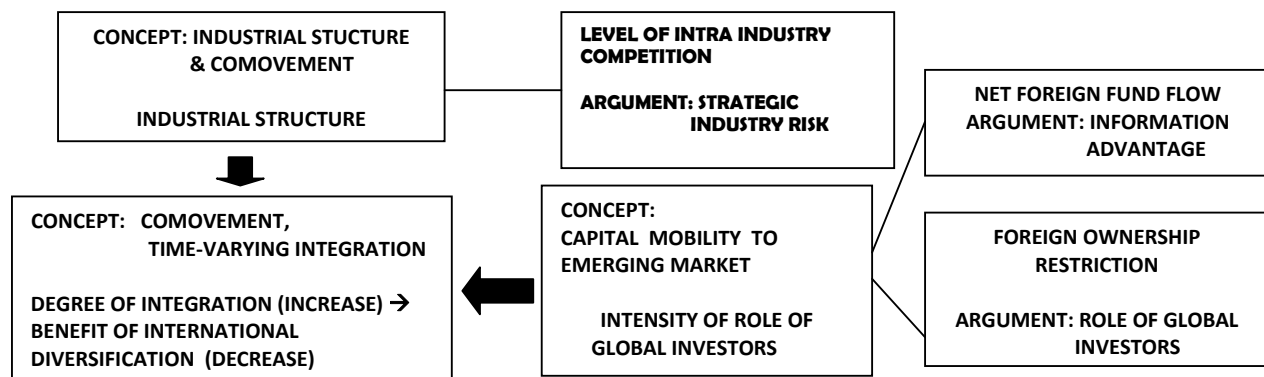


Figure 3.1
Conceptual Framework

Based on the conceptual framework above, then condition of the rising level of integration will be associated with lower benefits of international diversification. This is because when the market is integrated, global investors should consider the additional risk (global risk) due to increased correlations between a country's market index return with the return of the international index (comovement). Although there are indications that the degree of integration varies with time according to the concept of time-varying integration from Bekaert and Harvey (1995), but global investors need to look at inter industry diversification

is still hoping to reach the optimization of international diversification benefits. According to Ratner and Leal (2005) the optimization could be created by more focusing to the most prospective industries and make exit strategy toward the poor-performanced industries.

Level of integration and industrial structure associated with the concept of industrial structure and comovement of Roll (1992). Two characteristics of the capital market which has the same industry will increase the potential correlation of stock returns from both markets. In order to increase the effectiveness of diversification across industries, then the global investors do not dwell only on the condition of the concentration of an industry sector. They should begin to see the level of intra industry competition. Although an industry with a high level of competition will have a lot of strategic risk industry, but it still would be interesting if the industry is more members (players) which experienced an increase rating.

Increasing role of global investors exists through an increase in net foreign fund flows, then the level of capital market integration will increase [Froot and Ramadorai (2008)]. But when we realize the benefits of diversification is reduced, global investors will move out of emerging market investments. Transfer of investments that make the local authorities increase the level of foreign ownership restriction as this could destabilize the movement of market indexes. But when global investors want to increase the net foreign fund flow by reason of the level of integration has been dropped, then they need to have information superiority in the form of wider access to the exchanges that tend restrictive.

3.2. Hypothesis Development

3.2.1. The Difference of Capital Market Integration

Understanding about the different levels of integration in terms of intensity level the role of global investors and industry characteristics, first it is necessary described the condition of the capital market integration of such as: perfect integration, partial segmentation (mild segmentation), partial integration (mild integration) and perfect segmentation. Explanation of any conditions in addition to the capital markets will have implications for large and small difference in the international diversification benefits for global investors. However it also has implications for the difference between high and low integration. If the integration were high, they should focus on industrial diversification.

Forms of integration and segmentation above according to Errunza and Losq (1985) is caused by the existence of two classes of global investors are restricted and unrestricted, and the existence of two classes of securities to be eligible and ineligible. The definition of unrestricted global investors is investors who have no limits to purchase various securities in the capital market of a country. While restricted global investors are investors with the purchases limits of securities. This limitation arises because of the rule of the local regulator to protect domestic investors. The consequences are that restricted investors can only get eligible securities, while for unrestricted investors can gain eligible and ineligible securities. In perfect segmentation condition, then there are only restricted investors but not able to obtain eligible securities. This is because of the strict rules by regulator. As for the mild segmentation, there will only restricted investors but are still be able to have access to eligible securities for the strict rules of the game back loose. Another case for the mild integration, the number of unrestricted investors will be more dominant than the restricted but because of banded the strict rules then them only able to invest in eligible securities. And when conditions are perfect integration is achieved, then the global investors is able to diversify internationally in eligible and ineligible securities, but consequently the international diversification benefits will be minimum.

The existence of capital market liberalization in emerging markets such as quoted by Bekaert and Harvey (1995) make the entry barriers for global investors will be minimum.

This is because the increasingly high regulatory beliefs about the importance of the role of global investors in improving the performance of local exchange such as the strengthening of stock indexes and cooperation of global investors and local brokers. And then there was a condition shift from the perfect and mild segmentations to the mild and perfect integrations.

But Choi and Rajan (1997) found only a condition of mild integration in emerging capital markets as well as to mild segmentation. This indicates that the level of integration is no longer determined by the rules of regulatory limitation for global investors, but rather from a global investors's own motives for investing in a country or postpone it. The motive was based on the attractiveness of each stock are then realized in the form of positive and negative sentiment. The stronger the positive and negative sentiment that happens, the higher the level of integration that occurs. Because the context sentiment, it is usually temporary and it is in harmony with the concept of time-varying integration of Bekaert and Harvey (1995) that the level of integration can vary according to time. The difference is that global investors should be utilized to maximize the benefits of diversification while still looks to the condition in the level of integration.

After the monetary crisis, the U.S. investment flows become smaller in each ASEAN country and different for the Japanese investment flows. The existence of differences in U.S. and Japanese investment flows will indicate different levels of sentiment in each ASEAN country. According to Eun and Resnick (1994), this fact indicates different levels of participation of global investors which implies a difference in the level of integration of each ASEAN country. Based on this description, we propose a hypothesis:

First Hypothesis. *there is a difference in level of capital market integration among Indonesia (IDX), Malaysia (KLSE), Singapore (SGX), Thailand (SET), and Philippine (PSE).*

3.2.2. The Concept of Capital Market Integration Determinants

3.2.2.1. Concept of Entropy

In general, entropy is a measure of the amount of disorder in the system. High entropy means disorder is high while the low entropy reflects the regularity. According to the laws of thermodynamics, the higher the entropy will be more chaotic a system. In the science of industrial organization, entropy is often associated with concentration and competition [see Ng (1995)]. While the financial science, entropy has been used by Tu (1998) to test the integration of capital markets of Taiwan and the USA with entropy-based pricing (EBP) derived from Consumption CAPM (CCAPM) as discovered by Breeden (1979).

Ng (1995) suggests the concept of entropy to measure industry concentration is relevant because the entropy reflects the number of firms in the industry and the equity market. The essence of entropy in this context aims to measure the level of industry concentration through the information described as a probability distribution of market share. With so entropy can be interpreted as a measure of uncertainty associated company's market share. In her study Ng (1995) using the arithmetic mean value of the minimum and maximum entropy to measure the concentration index depends on the class boundaries, the number of players in one class and size class specific industries.

Because entropy is a measure of disorder, uncertainty and randomness in a system then according to Ng (1995) entropy will also be useful to measure industry competition. The argument of this is the entropy will vary according to shape the pattern of randomness or randomness that occurs. Entropy will depend on the number of firms in the industry and a growing number of companies in the industry will reflect the level of intra industry competition is getting higher and higher entropy. In addition the entropy depends on the

distribution of market share will increase competition and higher entropy will too. Entropy was inversely proportional to the concentration of industry, so that a high entropy will reflect the low concentration level and the competition will high.

Associated with Tu (1998), the concept of EBP is derived from the SDF model (Stochastic Discount Factors) proved the integration of Taiwan and the USA. Because of a component model for the function Langrangian EBP (L) is identical with the entropy index Ruefli (1990); $L = \sum_{j=1}^s p_j \ln(p_j/q_j) + \sum_{i=1}^n \lambda_i (-\sum_{j=1}^s p_j X_{ij}) + \lambda_0 (1 - \sum_{j=1}^s p_j)$, the concept of entropy will be relevant as a determinant of capital market integration.

3.2.2.2. The Concept of Intensity Role of Global Investors

Initially Bekaert and Harvey (2000) have not so justified the negative role of global investors that is as destabilizator (speculators). This is related to the phenomenon of increased capital market liberalization in ASEAN exchange authority on the belief that the role of investors in ASEAN investors will give added value to increase trading liquidity and market indexes. But with the Asian monetary crisis of 1997/1998 and 2007/2008 the global financial crisis is the perception of the authority of several exchanges began to change toward the role of global investors. They began to increase the level of resistance but in indirect form are generally in the form of protection against domestic investors. Conducting direct obstacle in the form of restrictions on current stock market is less relevant because of liberalization has done more than 20 years. Thus it may be clear that stock markets are opened for foreigner.

If there is more reason for global investors play a positive and negative role is always associated with efforts to secure the benefits of international diversification in emerging markets. As noted by Beaker and Harvey (1997), emerging market has two attractiveness for global investors such as high volatility and the market index return correlations are low. High volatility can be seen with the dynamics of the movement of market indexes in each ASEAN country. The existence of high volatility is attractive to global investors with short time horizons that do Covered Interest Arbitrage (CIA) which focuses on capital gains in exchange rate risk compensation. While the low correlation to attract global investors with long time horizon which generally as informed investors who seek the fundamental value.

Observing the negative and positive role of global investors, each authority should be familiar with it. Positive role can be seen from non negative NFFF (Net Foreign Fund Flow) what it means more global investors to channel funds to the local exchanges so that the local market indexes rose. However, because the essence of NFFF is hot money then possible occurrence of a negative NFFF. It can be recognized during the period of crisis in the form of falling market indexes. As a reaction to the negative role that the exchange authority can increase the level of protection for domestic investors. Level of protection is not just a tight capital controls, such as Malaysia but can be a limitation of series A and B shares in China's capital markets. But this protection must be well managed so that the positive role of global investors is not lost. This is because the status of emerging markets that depend on the intensity of the role of global investors can not be eliminated. Eventhough the regulators have an option to protect their domestic investors from the aggressive global investors.

3.2.3. Level of Intra Industry Competition and Degree of Capital Market Integration

Engwall (1973) and Ruefli (1990) conducted a study on the importance of industry structure for the company in terms of the dynamics of competition in an industry sector. They hold the view that the importance of analysis of competition between firms as measured by the entropy index in addition to knowing the position of superior and inferior inter-company now also be able to predict the position of superior and inferior among companies in the future. This is because technically the entropy index is calculated on the basis of time-series and use the ordinal scale (ranking) for example 1,2,3,4, ... n where for n indicates the number

of firms in the industry. Refer to Collins and Ruefli (1992) that is the nature of the dynamics of competition in entropy is derived from the information theory of Shannon (1948). As we know the information theory is related to disorder, uncertainty and randomness in a system.

Characteristic of the dynamics of competition between firms within the business environment can be brought to the dynamics of competition among the company's stock if the company is also related to listing on the stock market of a country. The use of entropy index in evaluating the performance ranking of companies listing on stock exchange will attract the attention of global investors because of the assessment ranking for this by using common ratio scale proved disappointing market participants as in the case of Enron and World dotcom and probably the fallen of Lehman Brother in 2008 due to global financial crisis.

When examined in the study of capital market integration, the author's knowledge no one has to use the entropy index Ruefli (1990) as a determinant factor. That has existed so far is proving the integration of capital markets in the context of industrial sectors such as studies Cavaglià, et al. (2000), Ratner and Leal (2005) and Antoniou, et al. (2007). They have a view of the context of sectoral integration is sufficient to provide a picture for global investors see the potential in each industry sector is viable or not as part of their portfolio of industrial diversification. However, when examined using the entropy index, the dynamic changes of each company in one industry sector will be more apparent, so too when they need full information about the general picture of the existing industrial sector in the capital market, then the numbers in the sub-component of entropy (lower, diagonal and upper entropy) is expected to be more objective in the assessment of strategic industry risk. Based on the description, the second hypothesis proposed is:

Second Hypothesis. *level of intra industry competition will influence toward degree of capital market integration in ASEAN countries.*

3.2.4. The Intensity of Role of Global investors and Capital Market Integration

Two arguments are used to explain the relationship between the two is the role of global investors and international financial integration through the concept UCIRP. Both of these arguments stem from a grand theory: international capital mobility. According to Sula and Willet (2009), with increasingly free flow of capital from developed country to a developing country as a result of liberalization and free trade, then physical activity, economics and finance of each country as if it had been fused. This happens because the enactment of a good standard in shape, size and price in each country who declare themselves as members of an economic and trade bloc, so that each country will declare readiness to open with each other and compete freely with each other. A manifestation of international capital mobility is greatest in the history of European economic integration which gave birth to the EEC (European Economic Community). On each member of the EEC (see now the EU) will occur in an efficient capital mobility, because member state have been open with each other and compete freely. As for non-members countries, then capital mobility occurs when common rules which would receive a reduction in restrictions.

At a higher level then the international capital mobility will reach the level of Optimum Currency Area (OCA), such as the formation of the Euro currency in 1999. When the context has led to the standardization of currencies like euro and dollar, the level of integration has led to a discussion of domain Interest Rate Parity (IRP). This is because the flow of capital that occurs because of differences in interest rates. Marston (1995), Bhatt and Virmani (2005) and Solnik and McLeavey (2009) state the interest rate differential may be relevant to CIRP (Covered Interest Rate Parity) and UCIRP (Uncovered Interest Rate Parity). CIRP related to international capital flows are not restricted tend to equate the nominal interest rate if they are tied up in a common currency (single). Seeing the EEC who already

have the Euro currency then it should be applied CIRP. However this is not easily realized because the EEC became the European Union has changed and although the Euro (except Pound Sterling) still exist but tend to be less bargaining power Euro against the U.S. dollar. Not easy to apply CIRP create the context of financial integration are discussed with UCIRP. UCIRP associated with unrestricted capital flows tend to equalize nominal interest rates.

An effort to cope with exchange rate risk, and then by taking into account differences in domestic interest rate (i_d) and abroad (i_f), make global investors are always looking for opportunities to do the CIA (Covered Interest Arbitrage). According to Marston (1995) and Solnik and Mcleavey (2009), the CIA can be done by purchasing foreign securities because of the condition of the forward discount. The phenomenon of the CIA in lines with the argument that there cointegration between international capital flows and exchange rates in addition to the interest rate even if only for the case of Indonesia. This is because global investors who bring different currency than the local currency on the one hand can be correlated with the movement of local currency and may also be correlated with stock market conditions. Based on a study of Dvorak (2005) and Aggarwal, et al. (2009) in Indonesia, the activity appeared to be particularly dominant global investors as domestic investors. But the unique despite the inferior performance of global investors in the short term, but were superior in the long run. Explanation of differences in the performance of global investors it is a phenomenon of the difference of information between global investors and domestic investors. Domestic investors are perceived to have the advantage of knowledge of local conditions of Indonesia, while foreign (global) investors are considered to have the experience and global network of brokerage as a form of information superiority. In order to become the most dominant, the investor needs to have a combination of local ownership and global capabilities of information brokerage.

Bekaert and Harvey (2000) state there are two roles of global investors in emerging markets. The first role relates to the actions of global investors in influencing the technical aspects of trading in a stock because it could potentially increase the liquidity, efficiency and value of related shares. While the role of the second act of a global investors to better obtain information superiority. But according to Bekaert and Harvey (2000) the role of negative impact if it is so global investors does not get the information advantages, then they will make a withdrawal. In larger-scale withdrawal of funds called the phenomenon of surge or sudden stop of capital flows [Sula and Willet (2009)]. This phenomenon triggers global investor's restrictions such in Malaysia since 1998. The restriction is conduct by increasing level of protection toward domestic investors. It is not only make limit purchase some series stock like as in China and Philippines but also increasing the tight control of capital outflow from Malaysia. Hence starting in 1999 Malaysia imposed this restriction by excluding its investable index from IFC. Based on previous description, we propose hypothesis as follow:

Third Hypothesis. *the intensity of the role of global investors tend to raise the degree of capital market integration in ASEAN countries.*

4. EMPIRICAL MODEL

In the test the first hypothesis (**H1**), we used ANOVA test with the F-test simultaneously for both country and industry level with the UCC and the DCC. To prove the Second Hypothesis (**H2**) and Third Hypothesis (**H3**), it would require empirical model to estimate the level of capital market integration. This empirical model will include the industry-level correlation $\rho(R_{ijt}, R_{wt})$ and two factors related determinants. The first determinant of $\rho(R_{ijt}, R_{wt})$ is the level of intra industry competition with proxy index of entropy ($E_{ij,t-1}$) and three supporting proxy DINDG_{ijt} namely, PGDP_{ij,t-1} and LNMCAPS_{ij,t-1}. Furthermore, the second determinant is the intensity of the role of global investors with the

two main proxy $NFFF_{jt}$ and FOR_{jt} and the alternative variable that is deviation IRP with the symbol $d(Fx)_{jt}$ functioning complement the testing analysis results of the Second and Third hypotheses are whether the integrated capital market condition, is still valid conditions of IRP (Interest Rate Parity). When the condition is integrated then the IRP should be held. Furthermore, empirical models for testing hypotheses two and three state as follows:

$$\rho(R_{ijt}, R_{wt}) = \alpha + \beta E_{ij,t-1} + \gamma DINDG_{ijt} + \theta PGDP_{ij,t-1} + \varsigma LNMCAPS_{ij,t-1} + \lambda NFFF_{jt} + \delta FOR_{jt} + \omega d(Fx)_{jt} + \varepsilon_{ijt} \quad (4.1)$$

notes:

$\rho(R_{ijt}, R_{wt})$	Correlation between R_{ijt} (industry return i in country j at period t) and R_{wt} (international indexes (world) return w at period t), where international index is MSCI. To calculate this correlation we used Pearson techniques which downloaded from spreadsheet. To count R_{ijt} we employ database as well as used to entropy index.
$E_{ij,t-1}$	Entropy index from industry i in country j at period t-1 which derived from Ruefli (1990) and Collins and Ruefli (1992). Process to calculate $E_{ij,t-1}$ is important to test the H2. $E_{ij,t-1}$ is calculated by adaptation formula of $H(S)_k = [\sum_i (\sum_j p_{i,j,k} \ln p_{i,j,k}) / q \ln q]$. To calculate $H(S)_k$ we conduct 3 steps: create a database for 10 GICS of 5 countries in ASEAN, to rank the firm in industry by net profit, to make a transition matrix (p_{ijk}).
$DINDG_{ijt}$	Global industry dummy ($D=1$) and regional ($D=0$). This variable is adapted from study of Faff and Mittoo (2003).
$PGDP_{ij,t-1}$	Proportion of GDP inter industry i in country j at t-1.
$LNMCAPS_{ij,t-1}$	Log natural of market capitalization industry i in country j at t-1.
$NFFF_{jt}$	Net Foreign Fund Flow in country j at period t.
FOR_{jt}	Foreign Ownership Restriction is one minus the ratio between MSCI Investable Index and MSCI Global Index in country j at period t [which could be written as follow: $FOR = 1 - (MSCI-II / MSCI-GI)$]. $FOR = 1$ means market is closed for global investors' participation while $FOR = 0$ means that market will open 100%.
$d(Fx)_{jt}$	Deviation of IRP in country j at period t. Formulation with $i_d - i_f$ or $(S_{t+1} - S_t) / S_t$. Code of Fx is <i>foreign exchange</i> in country j at period t and not an <i>identifier</i> .
α, β to ω	Intercept and coefficient for each independent variables.
ε_{ijt}	error (residual) for model 4.1 i.e. common factors beside $E_{ij,t-1}$, $DINDG_{ijt}$, $PGDP_{ij,t-1}$, $LNMCAPS_{ij,t-1}$, $NFFF_{jt}$, FOR_{jt} and $d(Fx)_{jt}$.

Reasons to use SURE: based on arguments Srivastava and Giles (1987) and Griffiths, et al. (1993) namely the existence of common factors which have influence on all dependent variables $\rho(R_{ijt}, R_{wt})$ but not in the system of equations simultaneously on the model 4.1. According to Bekaert, et al. (2011) common factors are derived from the nature of the ASEAN stock exchanges which have implicit barriers such as country of the local market, political risk and market regulation are often interrelated. When faced with a dynamic role of global investors will sometimes rule barring ownership of shares in a stock exchange authorities will be followed by the same rules in other exchanges.

SURE model in equation 4.1 consists of five equations y_1, y_2, y_3, y_4 and y_5 in its simplest form $y = X\beta + e$. Being if in the form of the matrix appears as follows:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{pmatrix} = \begin{pmatrix} X_1 & 0 & 0 & 0 & 0 \\ 0 & X_2 & 0 & 0 & 0 \\ 0 & 0 & X_3 & 0 & 0 \\ 0 & 0 & 0 & X_4 & 0 \\ 0 & 0 & 0 & 0 & X_5 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{pmatrix} \quad (4.2)$$

Where y_1 to y_5 will contain $\rho(R_{ijt}, R_{wt})$ on every stock in the ASEAN and X_1 to X_5 will contain the explanatory variables of $\rho(R_{ijt}, R_{wt})$ i.e. $E_{ij,t-1}$, $DINDG_{ijt}$, $PGDP_{ij,t-1}$, $LNMCAPS_{ij,t-1}$, $NFFF_{jt}$, FOR_{jt} and $d(Fx)_{jt}$. While β_1 to β_5 is a vector for every correlation coefficient $\rho(R_{ijt}, R_{wt})$ and e_1 to e_5 is an error vector for each equation. Model 4.2. above assumes that each error in the equation y_1 to y_5 will have a relationship or association called the contemporaneous correlation. Assumption (contemporaneous correlation) model 4.2 above can be formulated as follows:

$$\text{cov}(e_{it}, e_{jt}) = E[E_{it}, e_{jt}] = \sigma_{ij}, \text{ for } i, j = 1, 2, 3, 4, 5 \quad (4.3)$$

Dufour and Khalaf (2002) notes how the testing that is contemporaneous correlation:

H_0 : contemporaneous σ_{ij} covariance is equal to zero, for $i \neq j$
 H_a : at least one covariance is not equal to zero

In this statistical test used $\lambda = T^M \sum_{i=2}^{i-1} \sum_{j=1} r_{ij}^2$. Based on the test of H_0 , the value of λ (likelihood ratio) will have an asymptotic distribution of χ^2 with df that is formulated by $M(M-1)/2$, where M is the number of equations. H_0 is rejected if λ is greater than the critical value of χ^2 (df) in the table. Conversely, if λ is smaller than the critical value of χ^2 (df) in the table then H_0 will be accepted.

5. RESULTS & DISCUSSION

5.1. Hypothesis Testing Differences of Integration Level from 5 ASEAN Countries

The purpose of this first hypothesis testing is to verify the argument of Bekaert and Harvey (1995) about the the concept of time-varying world market integration and also the differences in conditions of capital market integration by Errunza, et al. (1992) are classified into four sections namely perfect integration, perfect segmentation, mild integration and mild segmentation. The importance of verifying the argument from Bekaert and Harvey (1995) is to know the extent of capital market integration will indeed change with time in the period of observation rather than longer periods of capital market liberalization in emerging markets decade of the 1990s and studies confirm the results of Hong (2005) and Baele and Inghelbrecht (2009) which supports Bekaert and Harvey (1995) with a further look at the context of ASEAN stock exchanges. While the importance of verifying the arguments is 4 types of capital market integration from Errunza, et al. (1992) is knows how far the different types of capital market integration will be responded by a global investors who will show the difference the benefits of international diversification. Umutlu, et al. (2010) state in the perfect integration condition, the benefits of international diversification will be minimum by reason of the volatility of capital markets will be extremely high and potentially reduce the benefits of international diversification. In order to get a maximum return then a global investors must be prepared to bear the maximum risk. Conversely when conditions are perfect segmentation, the benefits of international diversification will be maximum because the capital market volatility was low.

As the realization of the first hypothesis testing will be made in three stages namely: **first**, to test differences in the level of capital market integration in the respective ASEAN exchanges with the local exchange rate for R_{jt} . **Second**, to test differences in the level of capital market integration by each exchange in ASEAN at adjustment R_{jt} to the exchange rate of USD. This type of correlation used in phase 1 and phase 2 is the UCC (Unconditional Correlation). **Third**, do a comparison test result differences in the level of capital market integration with DCC (Dynamic Conditional Correlation). One of the implications of the

analysis of DCC is to check the function of the standardized residuals R_{jt} and R_{wt} to be seen how strong the effect of noise on the process for this transaction in any securities exchange. There are allegations that the Indonesian stock exchange practitioners' highest noise during the period 2006-2009. This can be caused due to the lack of FOR.

Based on Table 5.1, obtained an F-Test 6.28 is greater than F-table for 2.38 and significant at 1% significance level. So based on the testing table 5.2, **H1** is accepted which means that there are differences in the level of integration in ASEAN stock exchanges.

Table 5.1. Comparison Mean and Variance from Unconditional Correlation at Country Level [$\rho(R_{jt}, R_{wt})$] and F-test (Period 06-09)

Pairs of Unconditional Correlation	Mean	Variance	F-Test (F-table)
$\rho [R_{(STI)}, R_{(MSCI)}]$	0.325988	0.215142	28.643643*** (2.380743)
$\rho [R_{(KLCD)}, R_{(MSCI)}]$	0.181418	0.250112	
$\rho [R_{(IHSG)}, R_{(MSCI)}]$	-0.132551	0.252476	
$\rho [R_{(SET)}, R_{(MSCI)}]$	0.280908	0.237506	
$\rho [R_{(PSD)}, R_{(MSCI)}]$	0.078874	0.230304	

Source: Result Data Processing (2010) with *data analysis* EXCELL.

Table 5.2. Comparison Mean and Variance from UCC for each Industry Sector [$\rho(R_{ijt}, R_{wt})$] in ASEAN and F-test (Period 06-09)

Industry Sector (GICS Code)	Mean	Variance	F-Test (F-table)
Oil & Gas (10)	0.412215	0.107958	1.571152** (1.476896)
Basic Material (15)	0.561203	0.026617	
Industrial Goods (20)	0.570832	0.043517	
Services Goods (25)	0.338183	0.113971	
Consumer Goods (30)	0.340002	0.096903	
Health Care (35)	0.415157	0.068139	
Financial Institution (40a)	0.553977	0.087782	
Property and Real Estate (40b)	0.487187	0.089482	
Technology (45)	0.392070	0.080656	
Utilities & Telecommunications (50)	0.353708	0.105675	

Source: Result Data Processing (2010) with *data analysis* EXCELL.

In addition to testing for country level testing is also done for the industry level in Table 5.2. The purpose of testing to see whether the integration of capital markets is also happening within the industry level. Based on the results of testing, the F-test UCC > F-table and significant at a significance level of 5%, then **H1** is received again which means the existence of differences in sectoral capital market integration across the ASEAN stock exchanges. Like the unconditional correlations in local currency, then for adjustment also needs to be drawn for studying the behavior of the unconditional correlation of data in five ASEAN stock exchanges from 2006-2009. Unconditional correlation patterns USD from each ASEAN stock exchanges also have a wavy characteristic length but the upper range limit (positive correlation) and the lower limit (negative correlation) are almost equal. This correlation indicates unconditional USD is stationary according correlogram checking.

The difference of negative and positive correlation will be responded by global investors as part of important information for international diversification strategies. Based on table 5.3, the F-Test 41.9 > F-table for 2.38 and significant at 1% significance level, thus

H1 is accepted. It means that there are differences in the level of integration in the five ASEAN exchanges when the correlation was adjusted to the USD. This is consistent with Eun and Resnick (1994) which states USD does not affect the level of integration.

Table 5.3 Comparison Mean and Variance from Unconditional Correlation Adj. USD at Country Level [$\rho(R_{ijt}, R_{wt})$] and F-test (Period 06-09)

Pairs of Unconditional Correlation	Mean	Variance	F-Test (F-table)
$\rho [R_{(STI)}, R_{(MSCI)}]$	0.316763	0.224959	41.90549*** (2.380743)
$\rho [R_{(KLCI)}, R_{(MSCI)}]$	0.105576	0.256978	
$\rho [R_{(IHSG)}, R_{(MSCI)}]$	-0.259306	0.235022	
$\rho [R_{(SET)}, R_{(MSCI)}]$	0.230699	0.253271	
$\rho [R_{(PSD)}, R_{(MSCI)}]$	0.008485	0.238789	

Source: Result Data Processing (2010) with *data analysis* EXCELL.

In addition to testing for country level testing is also done for the industry level in Table 5.4. The purpose of testing to see whether differences in capital market integration also occurs at the industry level. Based on test results below, the F-test UCC > F-table and significant at a significance level of 5%, then **H1** accepted which means that there are differences among industry sectors of capital market integration in ASEAN stock exchanges.

Table 5.4. Comparison Mean and Variance from UCC Adj. USD for each Industry Sector [$\rho(R_{ijt}, R_{wt})$] in ASEAN and F-test (Period 06-09)

Sector Industry (GICS Code)	Mean	Variance	F-Test (F-table)
Oil & Gas (10)	0.422597	0.096186	1.591717** (1.476896)
Basic Material (15)	0.584893	0.029290	
Industrial Goods (20)	0.595375	0.052673	
Services Goods (25)	0.381971	0.103265	
Consumer Goods (30)	0.399815	0.116728	
Health Care (35)	0.428068	0.073245	
Financial Institution (40a)	0.566670	0.083217	
Property and Real Estate (40b)	0.549286	0.060559	
Technology (45)	0.418447	0.083797	
Utilities and Telecommunications (50)	0.389470	0.108095	

Source: Result Data Processing (2010) with *data analysis* EXCELL.

5.2. Unconditional Correlation (UCC) & Dynamic Conditional Correlation (DCC)

Comparison of UCC and DCC both local and USD currencies in each ASEAN exchanges done by selecting the year 2008. The reason is effect of the global financial crisis of 2007/2008 is relevant to prove the usefulness of the DCC correlation of stock returns in emerging markets and developed markets increased during the crisis period [Kuper and Lestano (2007) and Baele and Inghelbrecht (2009)]. In the crisis period we hope time-varying pattern of correlation which indicates the volatility clustering in the markets. This is because the increasing of comovement of stock market indices with developed market indices such as DJGI and MSCI. We use MSCI as proxy of international indices since it covers emerging market portfolio as well as developed market.

Based on table 5.5 panel A and B we find the significant difference in the UCC and DCC in Indonesia and Philippines. Specifically in Indonesia it is finding a negative correlation of UCC and positive form in the DCC and the highest level when testing t-test. It has implications for Indonesia's most integrated than the other 4 countries as the global

financial crisis. The next sequence in the integrated condition (t-test UCC and DCC) are the Philippines, Malaysia and Singapore. Surprisingly Thailand as source of monetary crisis of 1997/1998 shows no condition of integration difference. This is because none significantly of difference between UCC and DCC in Thailand which uses LOC and USD currencies.

Table 5.5 Comparison Mean and Variance from UCC and DCC in 5 ASEAN Stock Market and t-test for Sample Period of 2008

Panel A. Local Exchange Rate

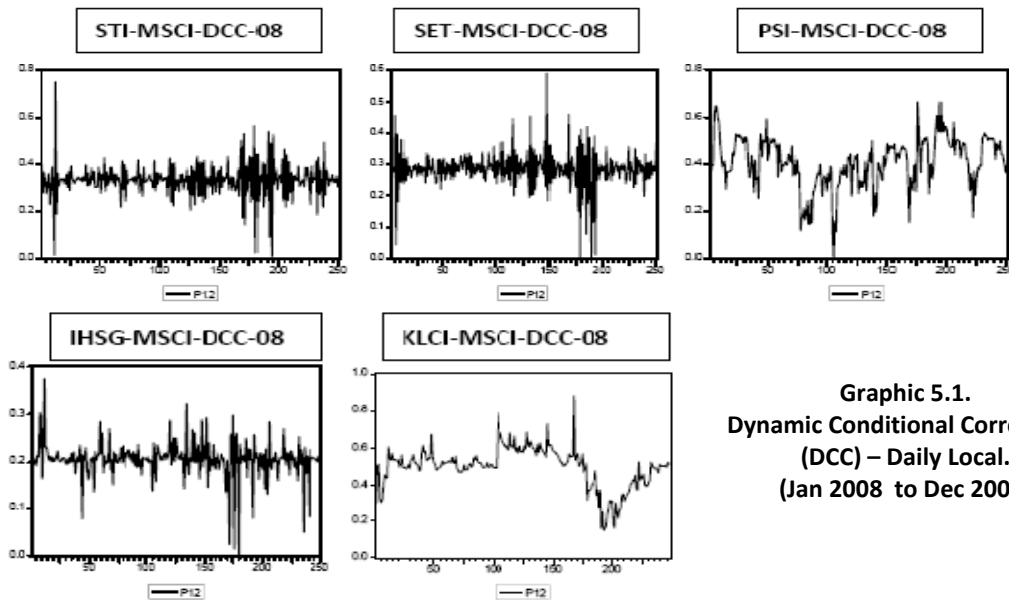
Pair of Correlations	Mean		Variance		t-test
	UCC	DCC	UCC	DCC	
$\rho [R_{(STD)}, R_{(MSCI)}]$	0.369026	0.346883	0.190948	0.004000	-0.354352
$\rho [R_{(KLCI)}, R_{(MSCI)}]$	0.275834	0.492558	0.249334	0.011665	2.967885***
$\rho [R_{(IHSG)}, R_{(MSCI)}]$	-0.193196	0.201142	0.227175	0.001652	5.669739***
$\rho [R_{(SET)}, R_{(MSCI)}]$	0.179888	0.283354	0.256349	0.005326	1.384783
$\rho [R_{(PSD)}, R_{(MSCI)}]$	0.156740	0.402135	0.265245	0.012394	3.171695***

Panel B. USD Exchange Rate

Pair of Correlations	Mean		Variance		t-test
	UCC	DCC	UCC	DCC	
$\rho [R_{(STD)}, R_{(MSCI)}]$	0.344968	0.458142	0.202459	0.025236	2.100281**
$\rho [R_{(KLCI)}, R_{(MSCI)}]$	0.192174	0.202646	0.262467	0.001700	0.144727
$\rho [R_{(IHSG)}, R_{(MSCI)}]$	-0.185120	0.191574	0.220538	0.001224	5.803634***
$\rho [R_{(SET)}, R_{(MSCI)}]$	0.191461	0.249483	0.266657	0.005285	0.795448
$\rho [R_{(PSD)}, R_{(MSCI)}]$	0.054285	0.291478	0.263307	0.012857	3.07117***

Source: Result Data Processing (2010) with *data analysis* EXCELL.

In order to verify the arguments increased correlations during the crisis, we apply it to DCC of 5 pairs of standardized residual return correlation for the STI-MSCI, SET-MSCI, PSI-MSCI, KLCI-MSCI and IHSG-MSCI. Based on the Graphic 5.1 for the DCC-Local currency static pattern was found for Singapore, Thailand and Indonesia. While the dynamic pattern was found for Philippines and Malaysia. DCC static patterns in Singapore and Thailand are consistent with the results of t-test in panel A of table 5.5 and also for the dynamic patterns of the Philippines and Malaysia. As for Indonesia it is found the DCC-static patterns of local currency that is not consistent with the t-test results in table 5.5.



Graphic 5.1.
Dynamic Conditional Correlation
(DCC) – Daily Local.
(Jan 2008 to Dec 2008)

In order to obtain to better ensure consistency of analysis results between tables 5.5 and Graphic 5.1, we have done in checking the effect of noise that occurs in table 5.6. In this case presented the type of process mean (ARIMA family) from Std. Resid. R_{jt} which became a major component of the DCC through the model GARCH (1,1).

Table 5.6 Comparison Function (Process) from Mean Std. Resid. R_{jt} for DCC-LOC in 5 ASEAN Stock Market [Example Period of 2008]

Pair of Dynamic Conditional Correlations	Mean Process from Std. Resid. R_{jt} (SR_ R_{jt})	Attribute of SR_ R_{jt}		
		SIC	AIC	D-W
SR_ R (STI)-SR_ R (MSCI)	AR(2) MA(2)	-5.00591	-5.09018	1.99403
SR_ R (KLCI)-SR_ R (MSCI)	AR(3) MA(3)	-5.69726	-5.78178	1.88016
SR_ R (IHSG)-SR_ R (MSCI)	AR(1) MA(1)	-4.64350	-4.75586	1.93376
SR_ R (SET)-SR_ R (MSCI)	AR(1) MA(1)	-5.01173	-5.09576	2.05874
SR_ R (PSI)-SR_ R (MSCI)	AR(2) MA(2)	-5.01155	-5.09582	1.67364

Based on Table 5.6 it was found that Singapore, Malaysia, Philippines and Thailand tend to have the same noise pattern. This can be determined by the mean equation process for Std. Resid. R_{jt} is ARIMA (p,d,q). While the Indonesian stock exchange has a mean equation process for Std. Resid. R_{jt} namely ARIMA (1,1,1). Mean equation process of Std. Resid. R_{jt} in Singapore and Philippines respectively at the ARIMA (2,1,2). Malaysia has a mean equation process of Std. Resid. R_{jt} namely ARIMA (3,1,3) and Thailand namely ARIMA (1,1,1). Corresponding argument of Kuper and Lestano (2007), the stock with the greatest noise effect is Thailand because of shorter lag order residual than other countries except Indonesia and most clustered pattern of DCC. Thus we imply Thailand will be more attracting the short term investors.

5.3. Hypothesis Testing for Determinants of Capital Market Integration

Listed in table 5.7 many independent variables have significantly effect toward level of capital market integration. It makes a great opportunity to accept the **H2** and **H3**. When analyzing the validity of SURE model, we get λ^2 -test of 39.393 (significant at 1% level). It is greater than λ^2 -table so that the assumption of residual contemporaneous correlation of SURE model is very satisfied. Thus we can use this model in hypothesis testing of **H2** and **H3**.

Table 5.7. Hypothesis Testing for Determinants of Capital Market Integration

INDEPENDENT VARIABLES	ASEAN COUNTRIES (OBSERVATION 2006-2009)				
	PHILIPPINES	THAILAND	MALAYSIA	SINGAPORE	INDONESIA
INTERCEPT	18.145 (2.03)**	-7.523 (-3.39)***	-50.646 (-2.46)**	11.408 (0.61)	8.194 (2.19)**
ENTROPI _(ij,t-1)	0.044 (0.60)	-2.145 (-2.47)**	0.027 (0.04)	-0.755 (-1.76)*	0.0059 (0.04)
DINDG _(ij,t)	0.151 (2.48)**	-0.188 (-2.19)**	-0.134 (-1.52)	0.088 (1.54)	0.151 (2.35)**
PGDP _(ij,t-1)	-0.001 (-0.01)	-0.746 (-1.76)*	0.527 (0.93)	-0.991 (-3.25)***	0.0138 (0.07)
LNMCAPS _(ij,t-1)	-0.0095 (-0.93)	0.048 (1.45)	-0.031 (-0.71)	0.074 (3.64)***	-0.003 (-0.27)
FOR _(j,t)	-22.812 (-2.04)**	14.138 (3.71)***	81.747 (2.50)**	-25.479 (-0.59)	-11.508 (-2.11)**
NFFF _(j,t)	0.0035 (0.24)	-0.011 (-1.54)	-0.134 (-1.85)*	0.0035 (0.38)	-0.585 (-2.20)**
DIRP _(j,t)	19.711 (2.96)***	-14.691 (-1.63)	-15.456 (-1.97)**	19.189 (0.70)	-37.082 (-1.86)*
R ²	0.33	0.55	0.27	0.44	0.31
λ^2 -test	39.393*** (Breusch-Pagan test of independence), <i>non iterated</i> SUR				
Notes:	***: significant at level 1% ** : significant at level 5% * : significant at level 10%		λ^2 -test is test for assumption of SUR i.e. contemporaneous correlation of residual (calculated by STATA 9.0)		

Empirical Model

$$\rho(R_{ijt}, R_{wt}) = \delta_0 + \delta_1 E_{(ij,t-1)} + \delta_2 DINDG_{(ij,t)} + \delta_3 PGDP_{(ij,t-1)} + \delta_4 LNMCAPS_{(ij,t-1)} + \delta_5 FOR_{(j,t)} + \delta_6 NFFF_{(j,t)} + \delta_7 DIRP_{(j,t)} + \varepsilon_{ijt}$$

Where:

- $\rho(R_{ijt}, R_{wt})$ = unconditional correlation between R_{ijt} and R_{wt} (by Pearson Correlation).
 $E_{(ij,t-1)}$ = level of intra industry competition of industry i at country j in period $t-1$.
 $DINDG_{(ij,t)}$ = dummy (category) industry i at country j in period t (global = 1, regional = 0).
 $PGDP_{(ij,t-1)}$ = proportion of GDP from industry sector i at country j in period $t-1$.
 $LNMCAPS_{(ij,t-1)}$ = log natural market capitalization industry sector i in country j $t-1$.
 $FOR_{(j,t)}$ = *foreign ownership restriction* at country j in period t .
 $NFFF_{(j,t)}$ = *net buying (selling)* stock by global investors at country j in period t .
 $DIRP_{(j,t)}$ = deviation of *Interest Rate Parity* at country j in period t .

5.3.1. Level of Intra Industry Competition

From testing of **H2** in table 5.7 it could be looked the significance of entropy as a proxy for the level of intra industry competition. Singapore and Thailand have significant entropy index, but it does not like Indonesia, Malaysia and Philippines. These results suggest **H2** is still accepted and indicate the relationship between the level of integration and the level of intra industry competition. Based on the analysis with lower entropy (*trend of firm as member industry move to the higher rank*) according to Ruefli (1990), the industrial good in Singapore and property & real estate in Thailand can be an option for global investors. This means that the effect of capital market integration condition that occurs in these two industries would not be a barrier for global investors to further enlarge the portfolio formation in both types of industry. This is because both industries have the advantage of being shown with more companies in the industry who have promoted compared with other. Especially global investors in Thailand and Singapore must look the dynamic of competition among firms in both industries in order to maintain the benefit of industrial diversification.

Meanwhile alternative proxies of entropy such as DINDG, PGDP and LNMCAPS give the different results. From the test results table 5.7, DINDG effectively influence the level of integration in the Philippines, Thailand and Indonesia. It supports **H2** again and consistent with Faff and Mittoo (2003). The implication is global investors can multiply the portfolio formation on the global GICS industry sectors while maintaining the ability to access information on a local brokerage in three countries. For testing with PGDP, we find PGDP effectively influence the level of integration in the Thailand and Singapore. So that **H2** is received again and this result support Carrieri, et al. (2008). It suggests beside dynamic of competition among firms in such industry, global investors should aware about the attractiveness of such industry in both countries respecting to Heckscher Ohlin Theorem.

Finally, LNMCAPS is effectively influence the level of integration only in Singapore. It indicates the acceptance of **H2** and consistent with study of Carrieri, et al. (2008). Thus it suggest Singapore is the best portfolio for global investors since the industry sector has closed to GICS and the dynamic of competition is most referred to entropy of Ruefli (1990).

5.3.2. Intensity of Role of Global Investors

Several countries experienced significant FOR except Singapore. But for NFFF only two significant countries, namely Indonesia and Malaysia. This findings is still accepted **H3**, although findings of NFFF and FOR asserts dominance in the FOR compared NFFF as a determinant of the level of integration. This means that the effect of market capitalization on the intuition does not change the still high protection against domestic investor's regulators. Worries about the negative role of global investors will be destabilize the monetary crisis in 1997/1998 still experience perceived clearly in the minds of the regulator to Thailand and Malaysia, which has a positive and significant FOR.

In contrast to the Philippines and Indonesian stock exchange which has a negative FOR indicates already low level of protection of domestic investors from the local regulator. Regulators aware of the potential global investors as a means to trigger the growth of liquidity traders market indexes because their actions are often the benchmark for local brokerage and local investors in improving the liquidity of the local trade. Increased liquidity in emerging stock markets such as Indonesia will be a positive consideration to other potential global investors to enter. With high liquidity and trade openness is still a big factor as well as coupled with a global network of brokerage owned by the global investors can optimally utilize both markets. In contrast to Malaysia and Thailand that the level of high protection against then domestic investors to global investors need to establish cooperation with a local brokerage for the effectiveness of penetration in these two markets.

In Philippines and Indonesia we find some mixed result about FOR and NFFF. This argument is simply the flow of international funds will easily flow into countries with low FOR. But what happened to Indonesia and the Philippines turned out to international fund flows are not so large that flowed. In contrast to Singapore with FOR was not significant but the direction of the positive NFFF. This underscores Singapore's still a reference to global investors in the flow of international investment funds. For the Philippines and Indonesia, the low flow of international funds can be caused due to global investors move some funds into capital markets, capital markets outside ASEAN is seen to be more prospective.

As a whole it is indicated the level of industrial countries and the global dominance of push factors is almost equal to the local pull factors is the finding that the Singapore stock exchange with the entropy index is always significant and the stock exchanges of Malaysia is the highest level of protection. That way, the SURE model as formulated in 4.1 and estimated in table 5.7 could bridge the divergent of Global Push Factors (GPF) and Local Pull Factors (LPF) as initiated by IRGISG and IFSG according to Pretorious (2002) and Hong (2005).

6. CONCLUDING REMARKS

6.1. Conclusion

The first finding is the difference in the level of integration of country and industry levels. It can be seen from the differences in unconditional correlations in the fifth ASEAN capital markets and differences in the unconditional correlation of tenth GICS industry. The findings of differences in the level of integration to the country level indicate a difference in the benefits of international diversification can be obtained by global investors. So that the benefits of international diversification are effective then global investors need to see the difference in the level of integration at the industry level of tenth GICS industry. Two findings support the opinion Carrieri, et. al. (2004) and Ratner and Leal (2005) that global investors can diversify the industries that have a low correlation.

Furthermore, when used dynamic conditional correlation (DCC), the difference in the level of country level integration is also still being discovered. This indicates the validity of the concept of time-varying integration of Bekaert and Harvey (1995) which makes global investors must constantly update its international diversification strategy in the 5th ASEAN equity markets due to the dynamics of correlation is often the case.

The second finding is the level of intra industry competition proved to affect the level of capital market integration in Singapore and Thailand. This indicates that global investors in determining the international diversification strategy does need to consider the factor of competition between firms within the industry especially has a tendency every company in the industry to increase ratings. The study by Collins and Ruefli (1992) need more attention by global investors due to higher ratings of each company in the industry means an increase above the target return of their international diversification strategy.

Then with proven entropy index as a proxy for the level of intra industry competition in influencing the level of integration supports the view of previous studies such as Tu (1998), Faff and Mittoo (2003), Carrieri, et al. (2004) and (2008), Dutt and Mihov (2008) that industry characteristic related to global investors to consider international diversification strategy. Specifically this study can be considered to provide the more adequate proxy for the level of intra industry competition in ASEAN. This is because many studies use Herfindahl index which actually is more appropriate to measure industry concentration. Being related to Tu (1998), then the findings supports the validity of the concept of entropy-based pricing, albeit in different forms i.e. not a non-parametric SDF (Stochastic Discount Factors).

The latest findings are that the intensity of the role of global investors was not so effective in raising the level of integration in ASEAN. It can be seen from no significance Net Foreign Fund Flow (NFFF) to the degree of capital market integration in ASEAN. These findings do not support the study of Richard (2005) and Froot and Ramadorai (2008) which state the integration of capital markets in ASEAN should be influenced by NFFF.

However, if associated with FOR then found a different thing. FOR proven several times significantly affect the degree of capital market integration. These findings support the results of the study FOR from Edison and Warnock (2003) on the persistence of the limits of regulators in several exchanges of the role of global investors. From the results FOR and NFFF, it is found clarity that there are limits to global investors make capital flow will be hampered. This can be justified because the data period in 2008 observation we associated with the global financial crisis the regulators in some markets will be more be more protective of domestic investors especially Malaysia.

Thus the general purpose of the study that is effectiveness of the of enforceability of capital market integration at the industry level has been reached. It can be seen the significance of differences in the level of integration of the industry level through the first findings of this study. The difference is not only to strengthening the industrial level integration studies from Carrieri, et al. (2004) and Ratner and Leal (2005) which based on

the industry level but also makes a major contributing of us to meet the linkages between capital market integration and the level of intra industry competition. Linking the two is when the ASEAN capital markets have integrated the inter industry diversification is more relevant than diversification across countries. In order to get the benefits of international diversification will be optimum, the global investors need to does a redefinition of inter industry diversification with more focus on the prospective industry.

6.2. Implications

With the achievement of two of the three objectives of this study that the different levels of integration through the correlation and the influence of intra industry competition level on the level of integration makes validation of SURE model that the author has proposed. SURE estimation model is actually a search of the correlation model of equation from Dutt and Mihov (2008) and Ferreira and Gama (2010) developed theoretical models of argumentation IAPT by Koutoulas and Kryzanowski (1994) and empirical models IAPT by Faff and Mittoo (2003). The main purpose correlation equation not to seek the relevant determinants of integration but also to determine the optimal correlation.

Corresponding argument of Chaieb and Errunza (2007) then the optimal correlation will show the best benefits of international diversification for global investors. In another form of the condition of the optimal correlation can be approximated by the concept of mild integration by Errunza and Losq (1985) and time-varying integration from Bekaert and Harvey (1995). According to these concepts, global investors need to be more flexible in managing its diversification strategy because there is no extreme condition for integration even though the era of liberalization has taken place two decades and more importantly, every exchange can be set FOR each as a form of indirect investment barriers. The main results of the corresponding author of the optimal correlation is shown by the magnitude of the correlation level both UCC and DCC but rather on the significance of entropy as the relevant local push factors in each market are more effective when global investors are more active role to foster partnerships with local brokerage.

For global investors are inter industry diversification options applying the entry and exit strategy from the Brooks and Negro (2006). Entry strategy carried out by taking the industrial sector with a lower entropy of the best and improves information superiority by networking at local brokerage. Exit strategy is done by rebalancing the portfolio in industry sectors with lower entropy is higher and networking with other local brokerage. For each regulation, the relevance of the entropy index shows that they need to more vigorously promote the industrial sector are listed on stock market in order to meet the ASEAN Stock Market Linkage (ASML) in 2015.

6.3. Possible Development

The next study should confirm Conditional Regime Switching (CRS) from Bekaert and Harvey (1995) so that it can play an effective role as a DCC from Engle (2002). This is because the CRS is able to capture the volatility asymmetry effect caused by DCC return data. Just to be able to implement CRS in SUR requires the SUR simulation model. Currently Zellner (inventor of the SUR) is being tried out by researchers SUR econometric simulation in Japan. The second thing is to link the integration of capital markets and noise traders like Gehrig's study (1993). Along with a growing finance research into the realm of behavioral finance and market microstructure is the role of global investors as value traders and noise traders in the integration of capital markets attract examined more deeply.

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Acknowledgement:

Many gratitude to the late, Mr. Bambang Hermanto, Ph.D [Senior Lecturer from Graduate School of Management, Faculty of Economics, University of Indonesia (UI)] who gives us much knowledge about the entropy of Ruefli (1990) as proxy of intra industry competition and the main determinant of the stock market integration in ASEAN stock market. Thus our study was more differentiated than the other studies. Many thanks too to helpful comments and constructive critics in order to increase the quality of this paper from the board reviewers at Graduate School of Management, Faculty of Economics UI such as Prof. Dr. Suroso, MBA and others such as Dr. Irwan Adi Ekaputra, Dr. Sugeng Purwanto, Zaafri A. Husodo, Ph.D and Viverita, Ph.D.